

Sonu Kumar

AI Engineer | LLM Applications, RAG & Agentic Systems

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SUMMARY

Computer Science graduate (B.Tech, 2026) who builds and deploys LLM applications, agentic workflows, and grounded **RAG systems**. Delivered three end-to-end projects with **Python, LangChain, LangGraph, MCP, Mistral AI, ChromaDB, PostgreSQL, and Docker**, covering tool routing, vector search, guardrails, persistent state, human-in-the-loop approval, API integration, and multilingual NLP.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C++
- **Generative AI & LLMs:** LLM Applications, RAG, Agentic Workflows, Multi-Agent Systems, Tool Calling, Prompt Engineering, Structured Outputs, NLP, Embeddings, Semantic Search, Machine Learning
- **Frameworks:** LangChain, LangGraph, FastAPI, Streamlit, Hugging Face Transformers
- **Models & APIs:** Mistral AI, Groq, Whisper, Sarvam AI, MCP, Tavily, AviationStack, OpenWeather, Alpha Vantage, API Integration
- **Databases & Vector Stores:** ChromaDB (Vector Database), PostgreSQL, SQLite, Sentence Transformers
- **Reliability & Deployment:** Guardrails, Human-in-the-Loop, LangGraph Interrupts, State Checkpointing, Automated Testing, LangSmith Tracing, Error Handling, Docker, Docker Compose, Git, GitHub, Render, Vercel

PROJECTS

Agentic Travel Planning System

2026

[Live Demo](#) | [GitHub](#)

Python · LangGraph · MCP · Groq · PostgreSQL · Streamlit · Tavily · AviationStack · OpenWeather

- Architected a supervisor-routed multi-agent workflow in **LangGraph** with typed shared state that extracts trip constraints and dynamically routes to flight, hotel, weather, and budget specialist agents.
- Integrated live flight, weather, and web-search data through **MCP** tools (AviationStack, OpenWeather, Tavily); added input guardrails and fallback handling so invalid requests and tool/API failures are handled gracefully.
- Implemented **human-in-the-loop** itinerary approval using LangGraph `interrupt()`, feedback-driven revision loops with a configurable revision cap, and PostgreSQL checkpointing to persist workflow state across pauses.

ResolveAI – Agentic Customer Support Resolution Copilot

2026

[Live Demo](#) | [GitHub](#)

Python · Streamlit · LangGraph · LangChain · Mistral AI · ChromaDB · SQLite · Docker · Render

- Engineered a conditional **LangGraph** workflow that classifies support tickets, retrieves policy evidence, drafts customer resolutions, and routes high-risk requests to a manager-approval step.
- Built a policy-grounded **RAG** pipeline with ChromaDB over support policies and historical tickets, grounding drafted resolutions in retrieved evidence; persisted tickets, resolutions, approvals, and activity logs in **SQLite**.
- Deployed a Streamlit application with **Docker Compose** on Render and wrote automated tests covering graph routing, policy decisions, database tools, and security logic.

AI Meeting Assistant

2026

[Live Demo](#) | [GitHub](#)

Python · LangChain · Mistral AI · Whisper · Sarvam AI · ChromaDB · Streamlit

- Built an end-to-end meeting and video intelligence pipeline that ingests YouTube links or local media and transcribes English with local Whisper and Hinglish with Sarvam AI.
- Generated transcript-grounded summaries, action items, decisions, open questions, sentiment analysis, formal minutes, and PDF reports using **LangChain** and **Mistral AI** workflows.
- Implemented transcript-grounded **RAG** with ChromaDB, rebuilding the vector store for each meeting to prevent stale context and cross-meeting answer contamination.

EDUCATION

Shivalik College of Engineering, Dehradun

Bachelor of Technology in Computer Science

Aug 2022 – July 2026

CGPA: 7.6/10.0

ACHIEVEMENTS & CERTIFICATIONS

- **Top 3 Academic Rank**, Shivalik College of Engineering, 2024
- Solved **200+ Data Structures and Algorithms problems** on [LeetCode](#)
- **AI Foundations Course Completion**, OpenAI Academy, 2026
- **Artificial Intelligence and Machine Learning**, Udemy, 2023